

How to create a Microsoft 365 Hybrid-Sync environment On Premises and M365 Cloud/ Syncing your On-Premises Users to the Cloud

How To Create a Microsoft 365 Hybrid Sync environment (On-Premises and M365 Cloud)

And Syncing your On-Premises Users to the Cloud

Product Name: Microsoft Exchange, Azure Active Directory, Entra ID, Microsoft Entra Connect, Microsoft 365

Version: Entra Connect Sync 2.6.1.0 / Exchange 2019 CU15 / Windows Server 2022 DataCenter

Phase: Deployment

Date:

Apr 7, 2026

Overview

Microsoft 365 (M365) Microsoft 365 is a cloud-powered subscription service from Microsoft that bundles premium productivity apps with cloud services and advanced

security features

Email → Outlook/Exchange

File storage → OneDrive

Collaboration → Teams, SharePoint

Apps → Word, Excel, PowerPoint

Security - Microsoft Defender

In short:



It helps you *work, communicate, and collaborate.*

Microsoft Entra (formerly Azure AD)

Microsoft Entra A *cloud identity and access management platform* that controls *who can access what*. Key features:

User sign-in & authentication

Single Sign-On (SSO)

Multi-Factor Authentication (MFA)

Conditional Access policies

Identity protection

Product Objectives



"How To Create a Microsoft 365 Hybrid environment (On-Premises and M365 Cloud)"

This document walks IT Professionals and Businesses through building a Microsoft Hybrid (On-Prem+Cloud) environment from scratch—installing Windows Server, configuring Active Directory, and setting up Exchange Email Services, Setting up Exchange Online/Entra AD communication endpoints to the Internet and your Internal Business.



This helps users avoid guesswork and follow a proven process.

2. Reduces errors and troubleshooting time You include important checks (like DNS setup, prerequisites, and PowerShell commands).



This helps users *avoid common mistakes* that could break things later.

3. Helps users achieve the end goal

The goal of your product/guide is to:

- Build an AD environment

- Deploy Exchange

- Create mailboxes and send email

Step: 1 Hybrid-join Microsoft Active Directory with Microsoft Entra ID 365

Hybrid Connect Entra Connect and Cloud Sync

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment/deploy-hybrid>

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment-prerequisites>

<https://learn.microsoft.com/en-us/microsoft-365/admin/setup/add-domain>

<https://learn.microsoft.com/en-us/microsoft-365/enterprise/urls-and-ip-address-ranges?view=o365-worldwide>

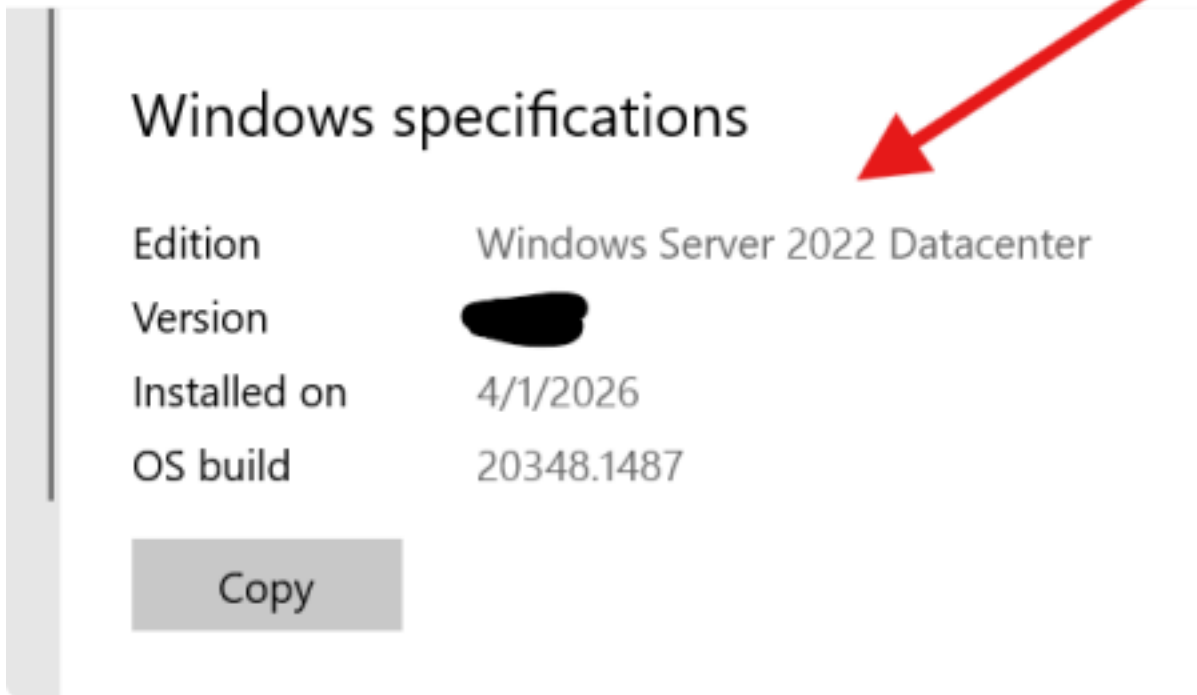
<https://learn.microsoft.com/en-us/microsoft-365/admin/get-help-with>

[domains/create-dns-records-at-any-dns-hosting-provider?view=o365-worldwide&tabs=domain-connect](https://learn.microsoft.com/en-us/microsoft-365/admin/get-help-with-domains/create-dns-records-at-any-dns-hosting-provider?view=o365-worldwide&tabs=domain-connect)

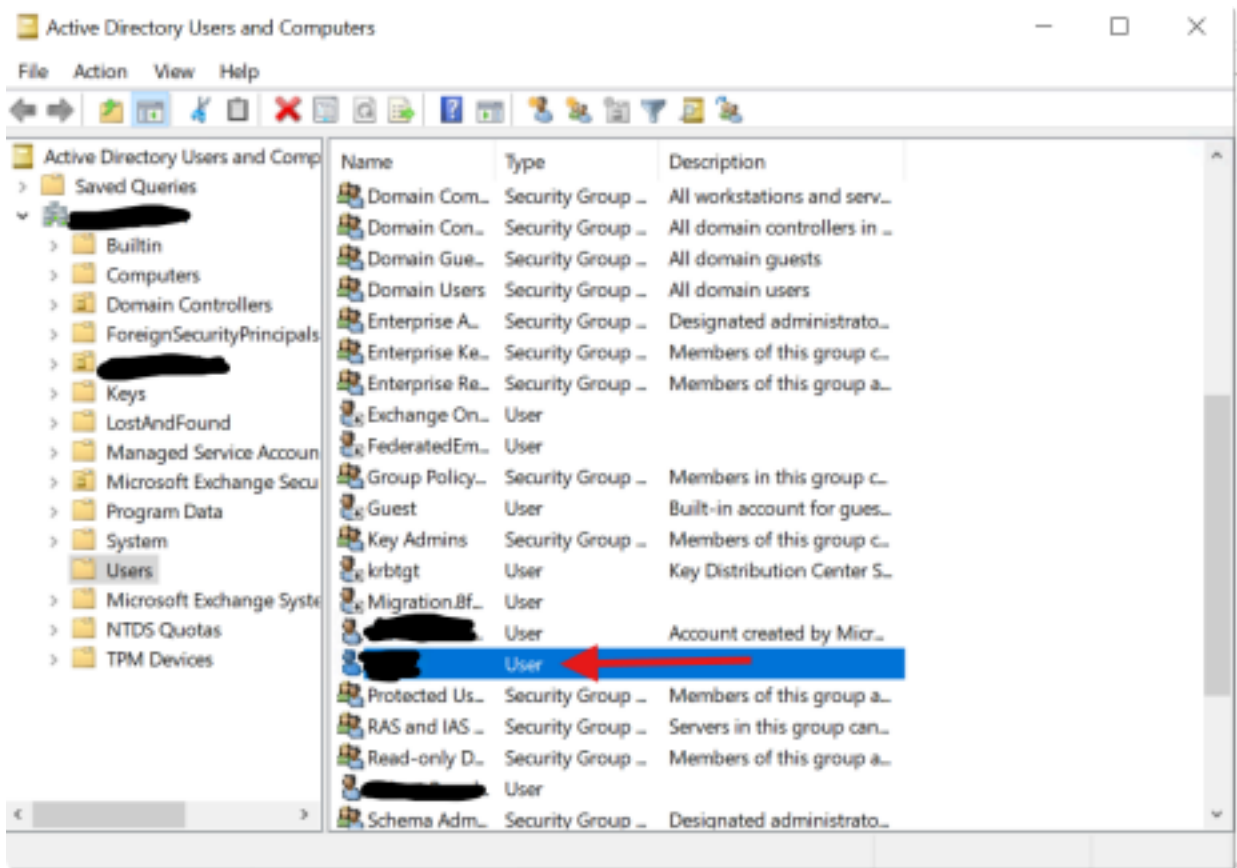
First we need to check that Pre-requisites are met -

From your **On-Premise Server** - ensure you have a **User/Admin** assigned : **Enterprise Admin Role** credentials for on-prem Active Directory and a **Licensed Windows Server** (2025,2022,2019) Standard or Datacenter Editions.

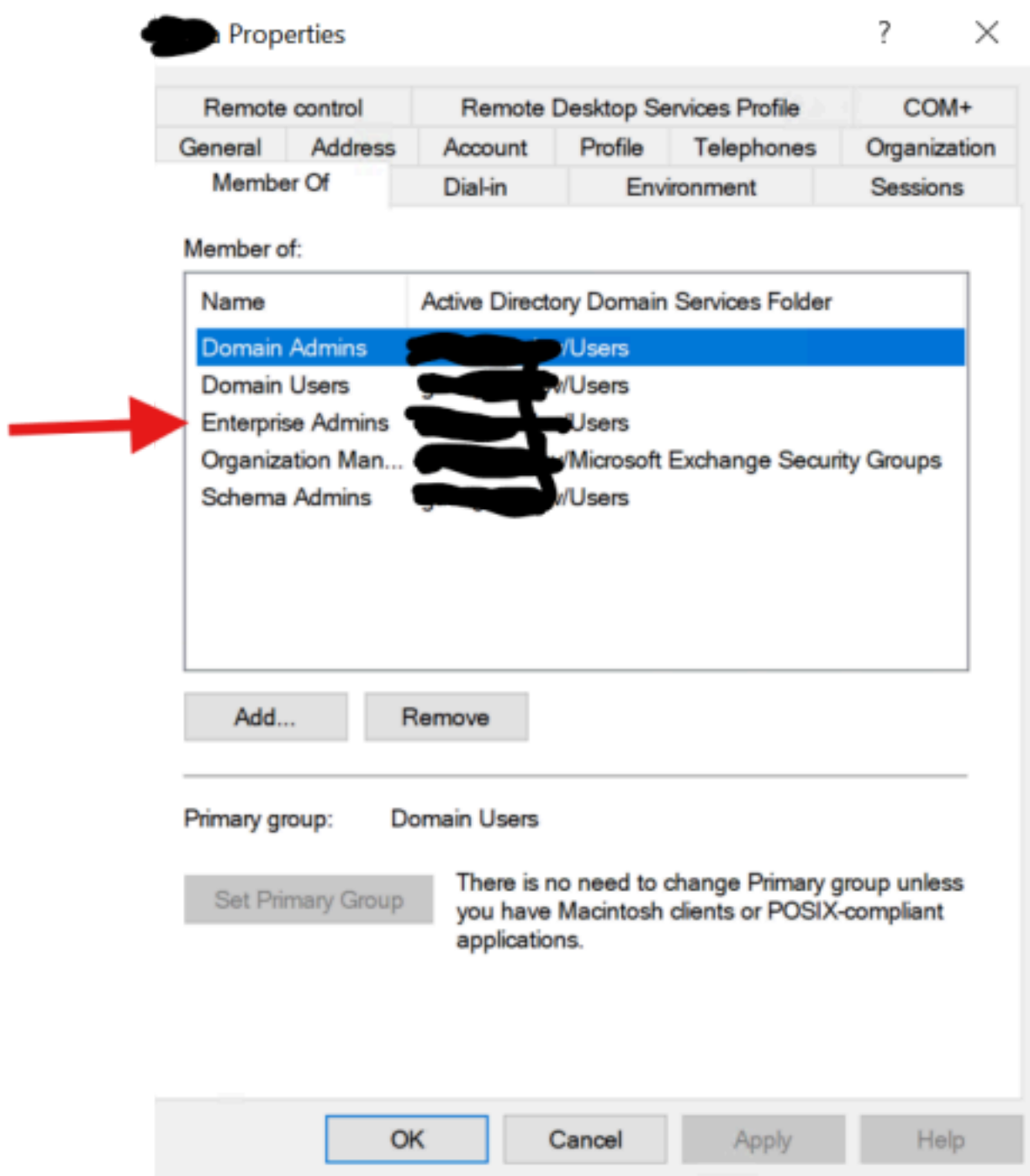
From your M365 Cloud - ensure you have a User/Admin with the Hybrid Identity Administrator Role for Entra Identity . (Images Shown Below)



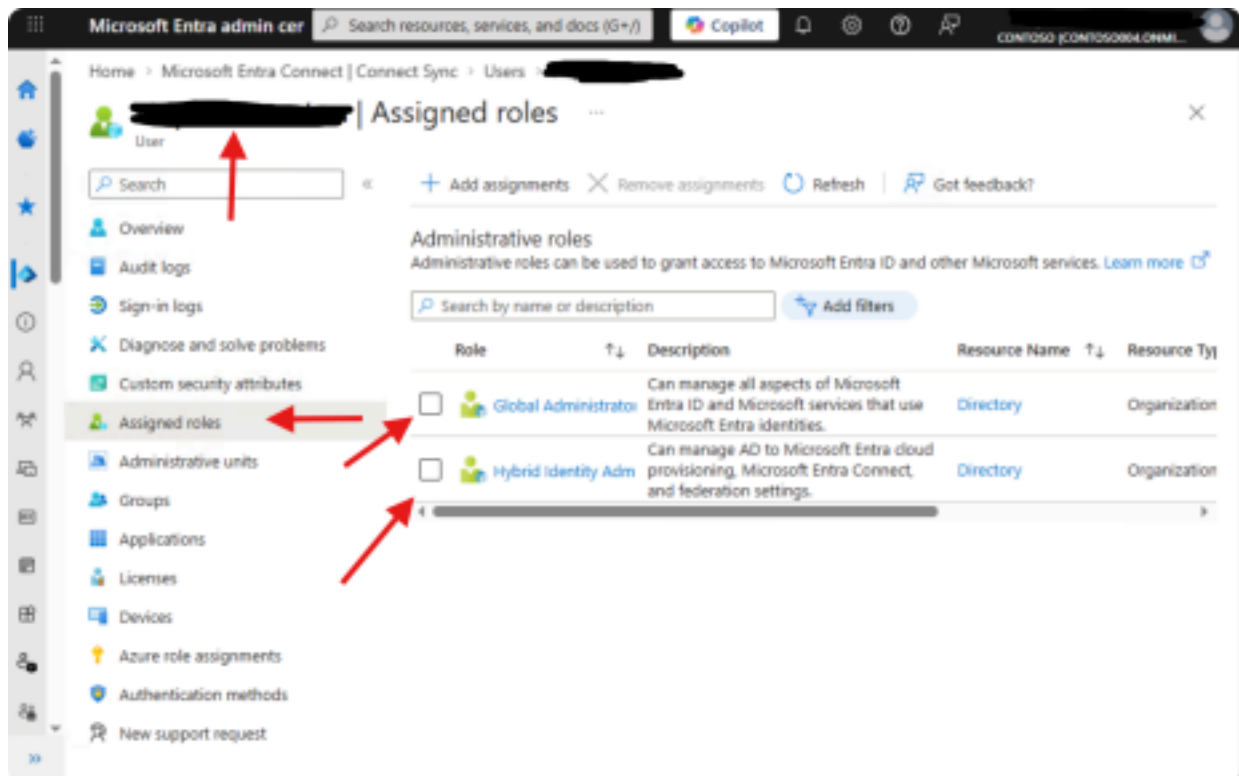
Add a caption...



Add a caption...



Add a caption...



Add a caption...

This URL Explains the Requirement for your Hybrid

Deployment <https://learn.microsoft.com/en-us/exchange/hybrid-deployment/deploy-hybrid>

Exchange 2016 and newer: At least one Mailbox server.

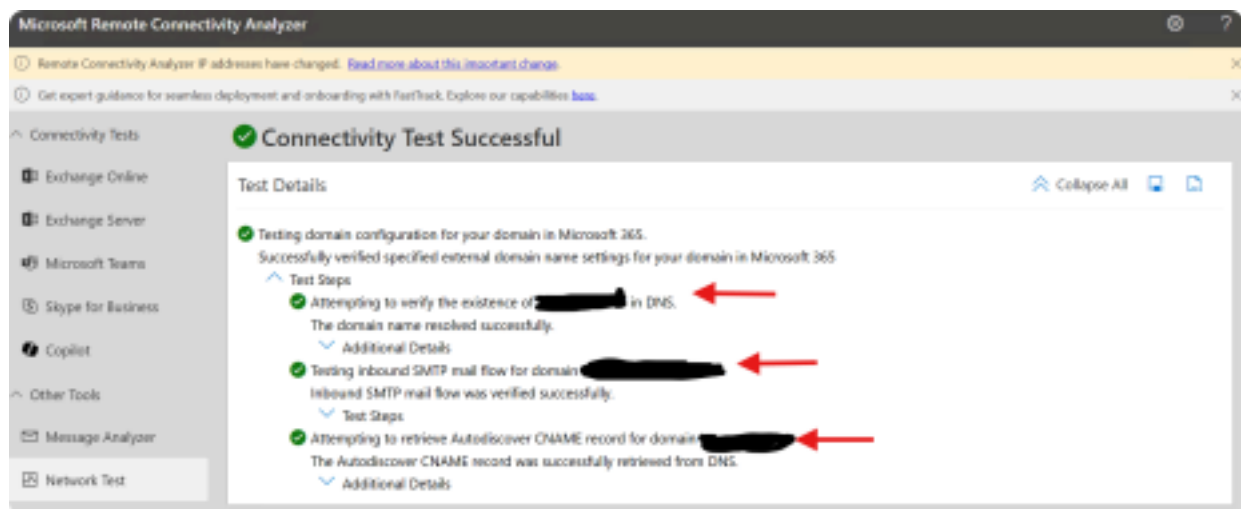
Active Directory synchronization: Deploy the Microsoft Entra Connect or cloud sync tool to enable Active Directory synchronization with your on premises organization.

Custom domains: Register any custom domains you want to use in your hybrid deployment with Microsoft 365 or Office 365. You can do this by using the Microsoft 365 portal, or by optionally configuring **Active Directory Federation Services (AD FS)** in your on-premises organization.

Make sure you have actively allow ports to communicate between on-prem and M365 (Port 25 SMTP, 143/992 IMAP 80/443 HTTP/HTTPS, 587 SMB and 53 DNS). Active Directory synchronization between the on-premises organization and the cloud, which is performed every 30 minutes by a server running Microsoft Entra Connect, is a requirement for configuring a hybrid deployment.

And lastly, perform a check for your HCW readiness with Remote Connectivity Analyzer

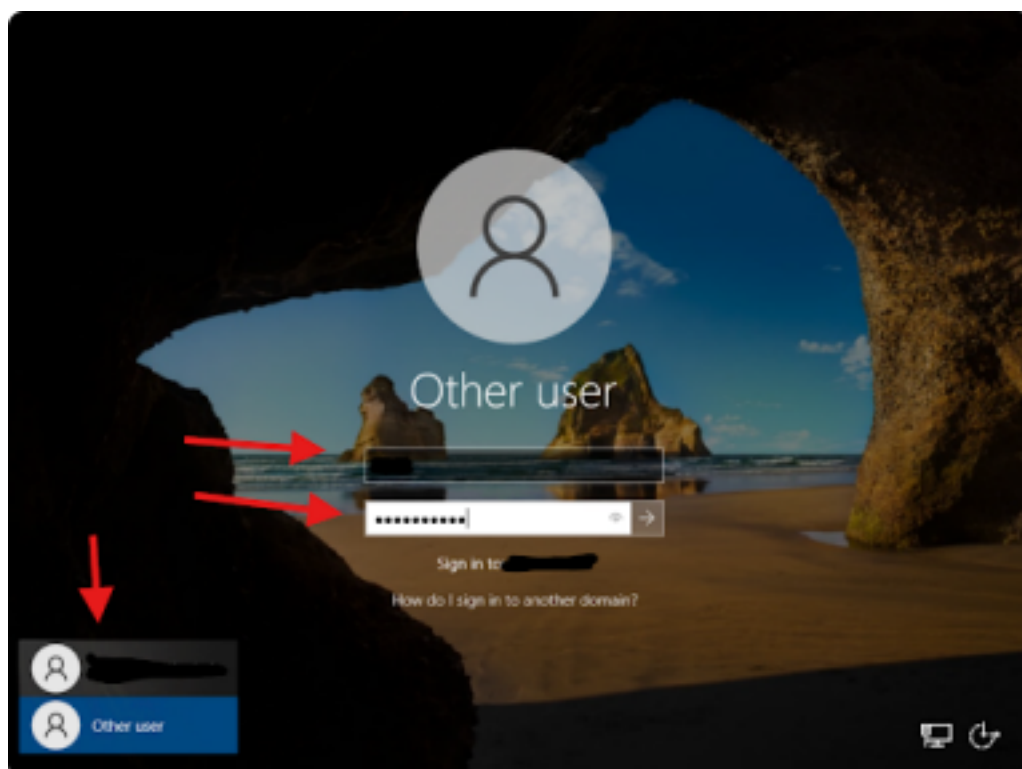
The Microsoft Remote Connectivity Analyzer tool checks the external connectivity of your on-premises Exchange organization and makes sure that you're ready to configure your hybrid deployment. <https://testconnectivity.microsoft.com/>



Section 2: Installing Entra Connect for Azure AD On-Premise Sync and Cloud Entra ID Sync

Part 1: Installing Azure AD Connect

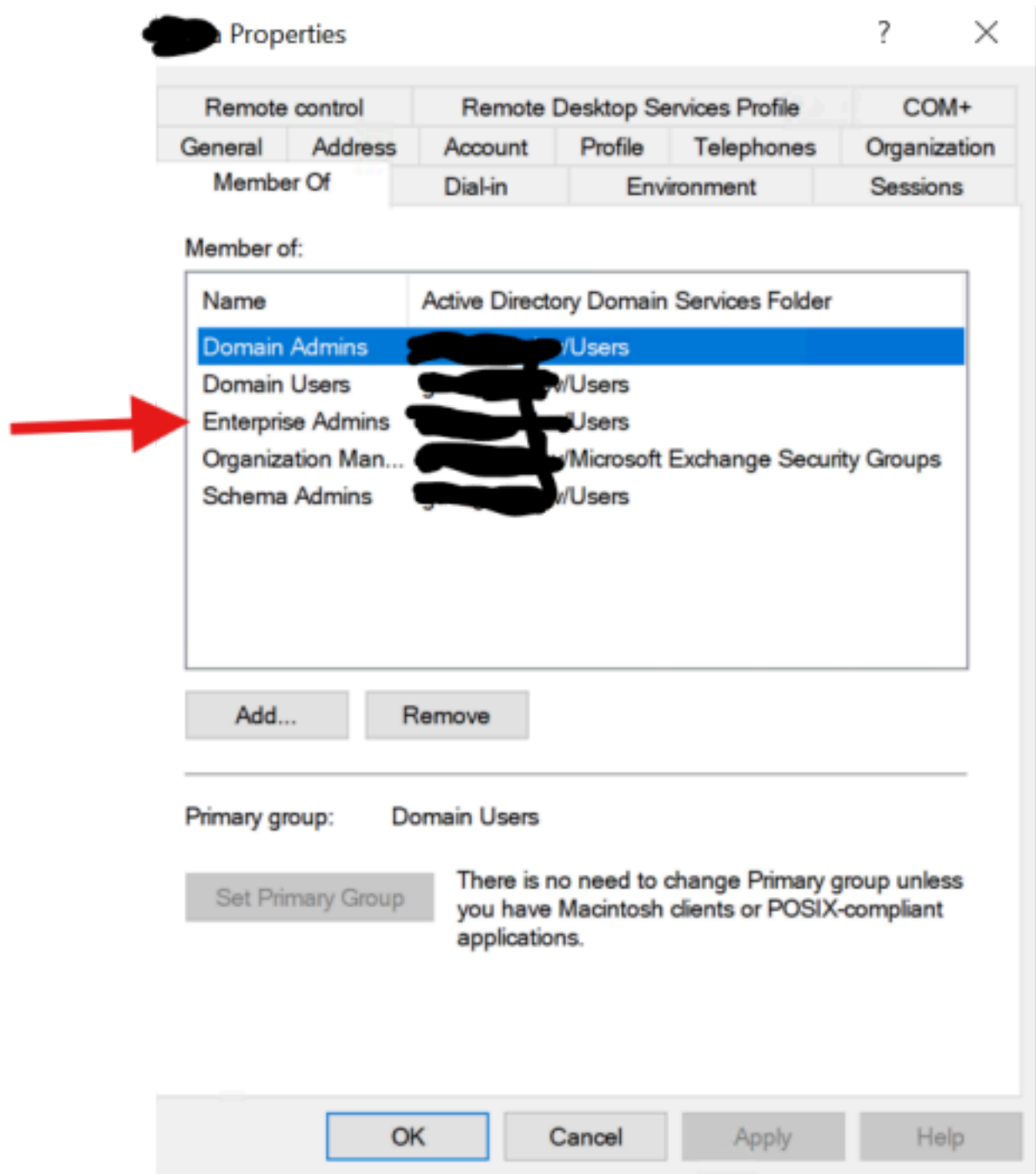
First Sign-in to your On-Prem Active Directory With your "Enterprise Admin" or "Domain Admin" account



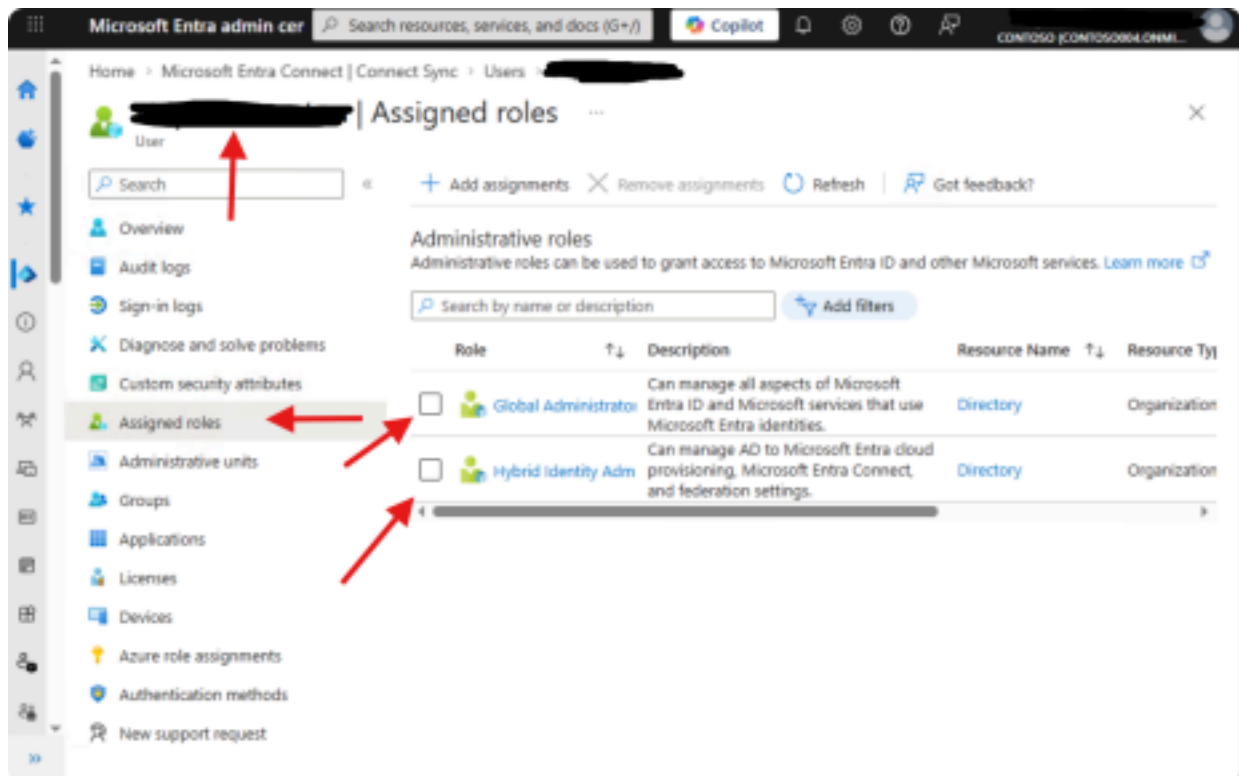
Add a caption...

Second go to your Microsoft Entra admin Center Sign-in with your Global

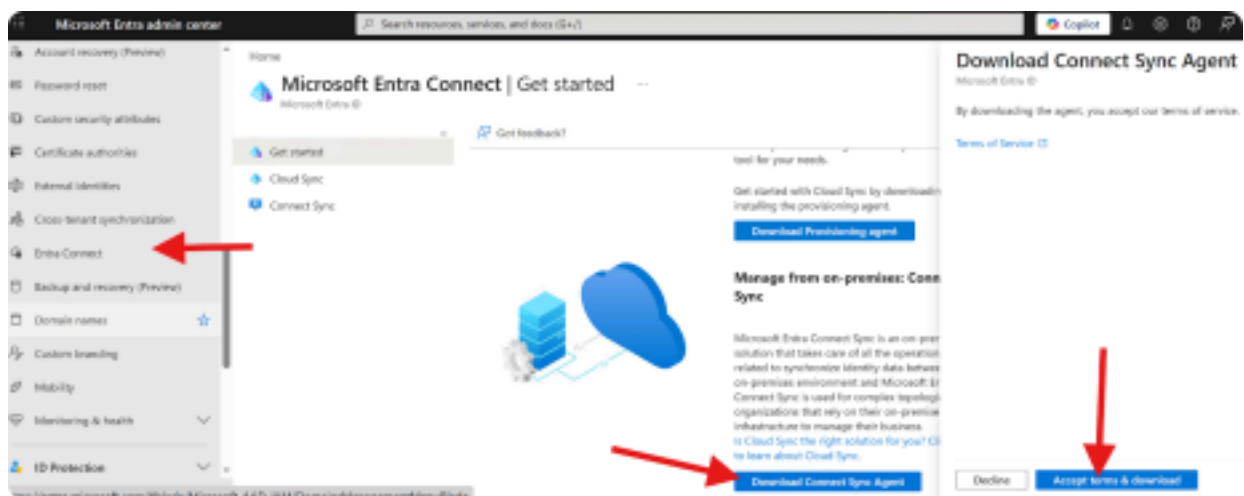
Administrator account - Go to Entra ID Tab- Select “Entra Connect” for On-prem Syncing Click the “Download Connect Sync Agent And Accept the Terms and Conditions.



Add a caption...



Add a caption...

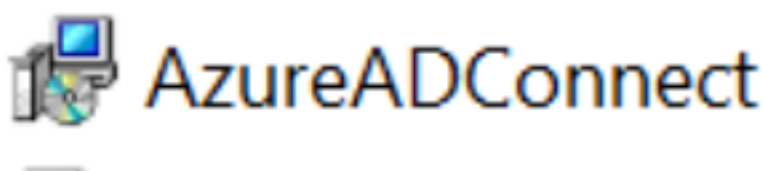


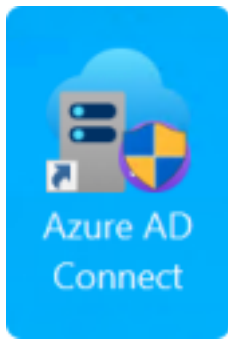
Add a caption...

Since we Already Have an **Exchange** and **Active directory** we will Go through the **"Express Settings"** unless any Customization is needed.

IMPORTANT

- BE SURE YOU KNOW THE CREDENTIALS FOR YOUR ON-PREM ENTERPRISE ADMIN / AND YOUR CLOUD ADMIN (Hybrid Identity Admin/Domain Name Admin")
Start the Install of "Entra Connect Sync" as Admin and the .Msi will begin,





Add a caption...

Connect Your M365 Entra Hybrid Identity Administrator to the "Username" Area Microsoft MFA will Pop-up for you to sign-in the Your Hybrid Admins

Credentials.



Add a caption...

Next, Enter your **AD DS Enterprise administrator** Credentials to Sync both **On premises Exchange 2019** and **Microsoft Entra 365** Administrator Accounts.



Add a caption...

Once you have Reached the "Ready To Configure" Panel Be sure to checkmark Both "Start the Synchronization Process" And "Exchange Hybrid Deployment"

*Enables synchronization of specific Exchange attributes between on-premises Active Directory and Exchange Online (Office 365). It is crucial for co-existence, allowing shared address lists, calendar free/busy sharing, and proper mailbox management during or after migrations.



Add a caption...

Once you've Press "Install" wait for the **Entra Connect Sync** to Update Synchronization Rules/Data and prepare your **On-Premises** and **Microsoft 365 Cloud Tenant** to build a communication to sync **Attributes** and **User/Computer Identities**.



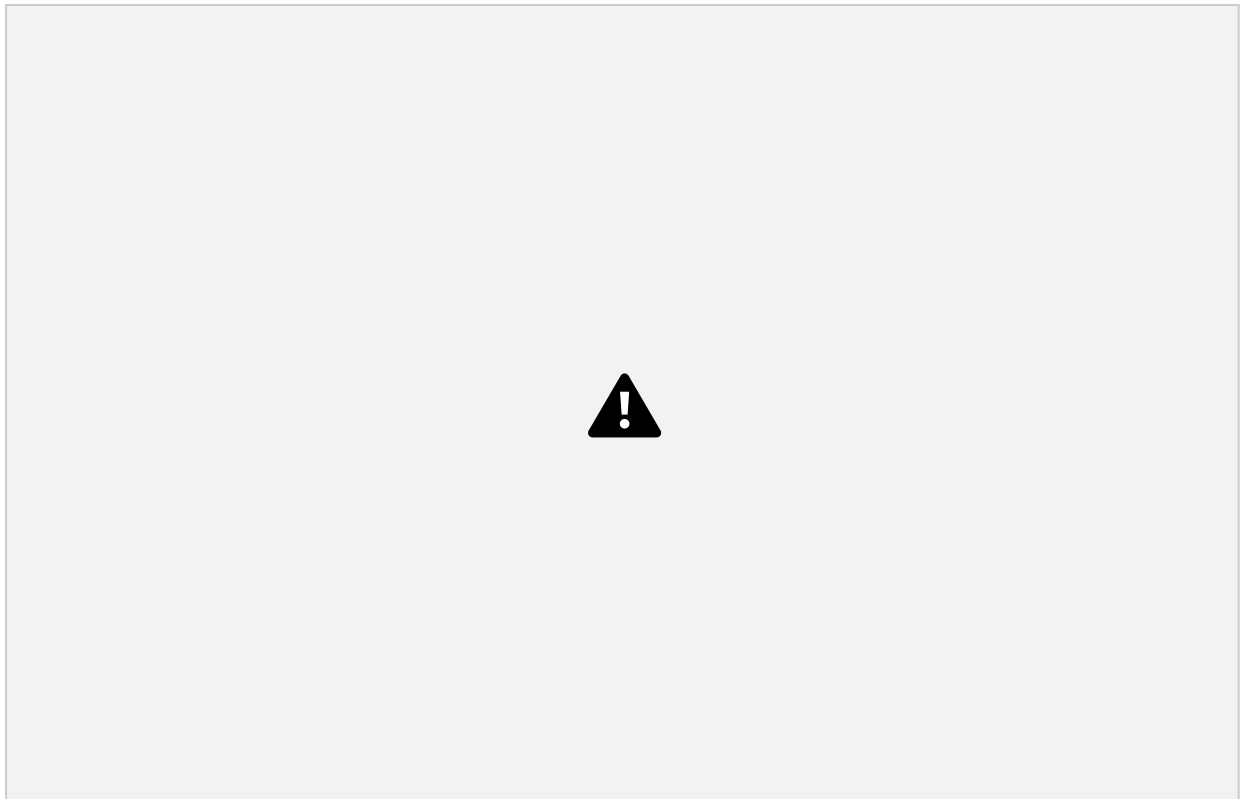
Add a caption...

The Wizard has now created a communicated Hand-shake between your **Azure AD/ Entra ID Objects, Identities, and Users/Computers**.



Add a caption...

Return back to your “**M365 Entra admin Center**” Then go to “**Entra Connect**”, Then “**Cloud Sync**”

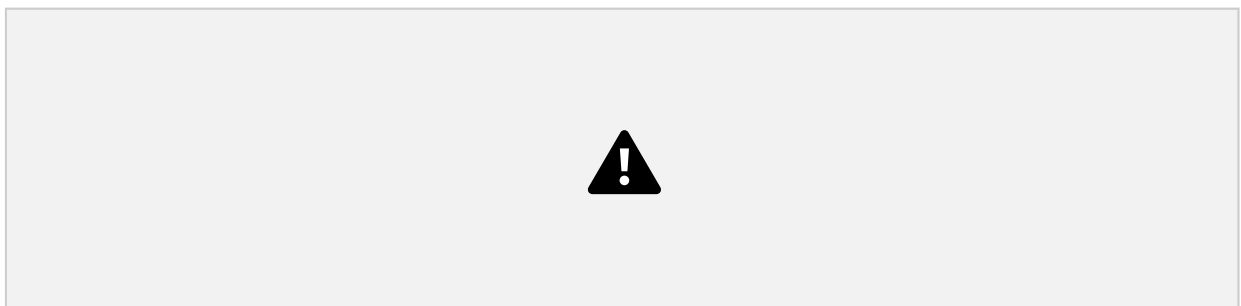


Add a caption...

Under "Cloud Sync | Configurations" we now see our first sync connection to make a "Hybrid Configuration "

WEVE MADE OUR FIRST CONNECTION FROM CLOUD TO ON-PREMISES

Microsoft ENTRA CLOUD SYNC



Add a caption...

Now Verify that the “Cloud Sync Agent has been installed in Entra from : “**Entra**

connect” - Cloud Sync - “Agents” and “All Registered Agents to This Tenant” That would be a YES



Add a caption...

And On your **Windows SERVER** Services “**Service.msc**” You will find the New
“AD Connect Provisioning Agent and Agent Updater”

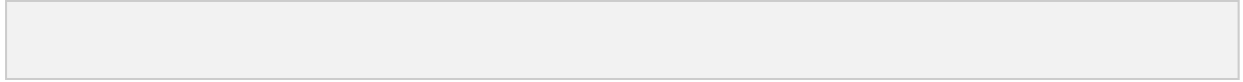
Service.msc



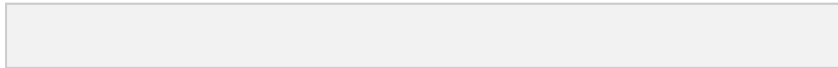
Add a caption...

[Verify your provision Agent version](#)

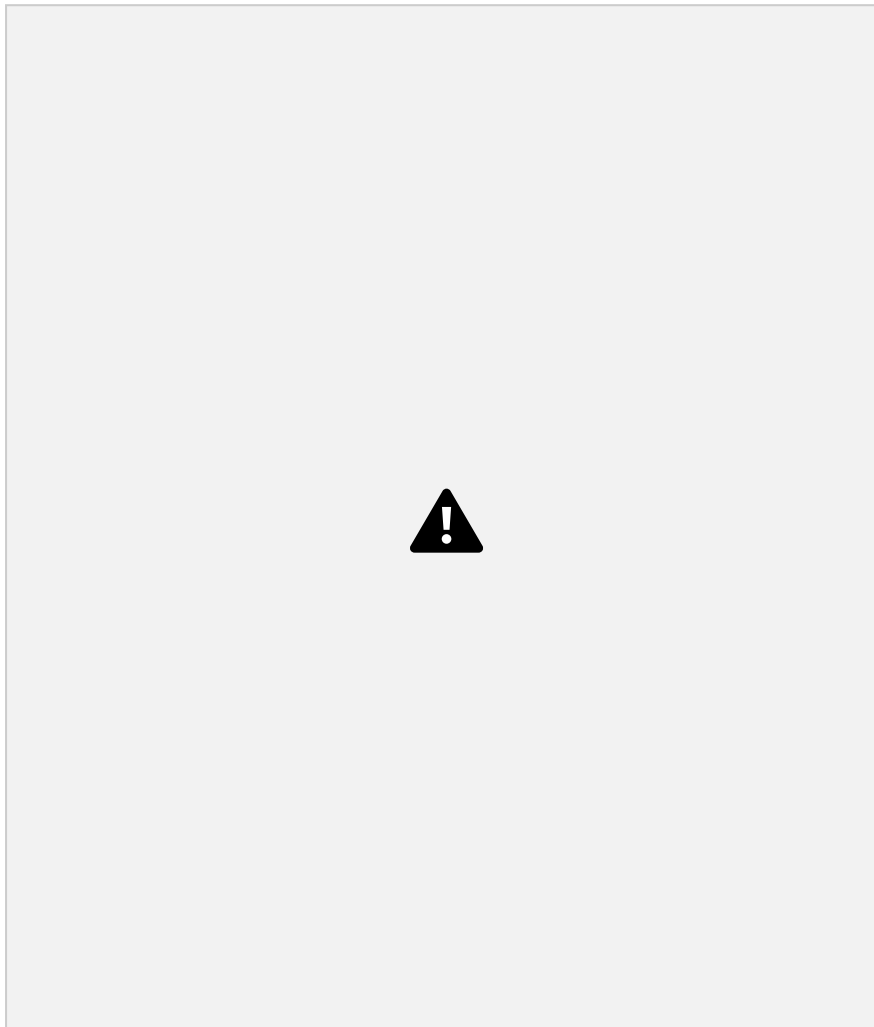
Go to C:\Program Files\Microsoft Azure AD Connect Provisioning Agent.
Right-click AADConnectProvisioningAgent.exe and select Properties. Select
the Details tab. The version number appears next to the product version.



Add a caption...



Add a caption...



Add a caption...



GOOD TO GO!!!!

IMPORTANT: For the Next Step Adding “New Configurations” From On-Premises AD to Entra ID. Be sure to "Allow" several needed addresses and ports within your On-Prem

Server BEFORE continuing your Hybrid Configuration. Within your On-Premises Server go to the Windows Search bar and Type “Internet options”. Under “Security” add the following URL’s to your “Trusted Sites Page” in “Internet Properties”

- *. [msapproxy.net](#) . / .servicebus.windows.net
- [login.windows.net](#) / [login.microsoftonline.com](#) (These 2 for initial registration) -
[mscrl.microsoft.com:80](#) , [crl.microsoft.com:80](#) , [ocsp.msocsp.com:80](#)
[www.microsoft.com:80](#)



Add a caption...



Add a caption...



Add a caption...

Once you've finished those URL additions go back to "**Entra Connect**" in the "**Entra Admin Center**" configure the Last of the "**Cloud Sync**"

Now within **Entra Connect** from **Entra Admin center** - select "**Cloud Sync**",
Select the "+ New-Configuration" Drop-down List and Click on "AD to
Microsoft ID sync"



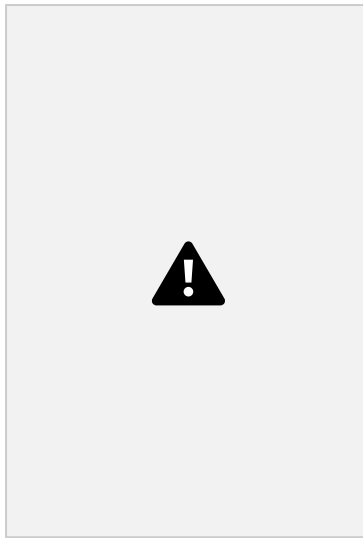
Add a caption...

Here you will see your On-premises Provisioning Agent install Link - Ensure your **Active Directory** Server is "**Active**" in the "**Agents configured for**" here by **Entra ID** before Downloading the Agent

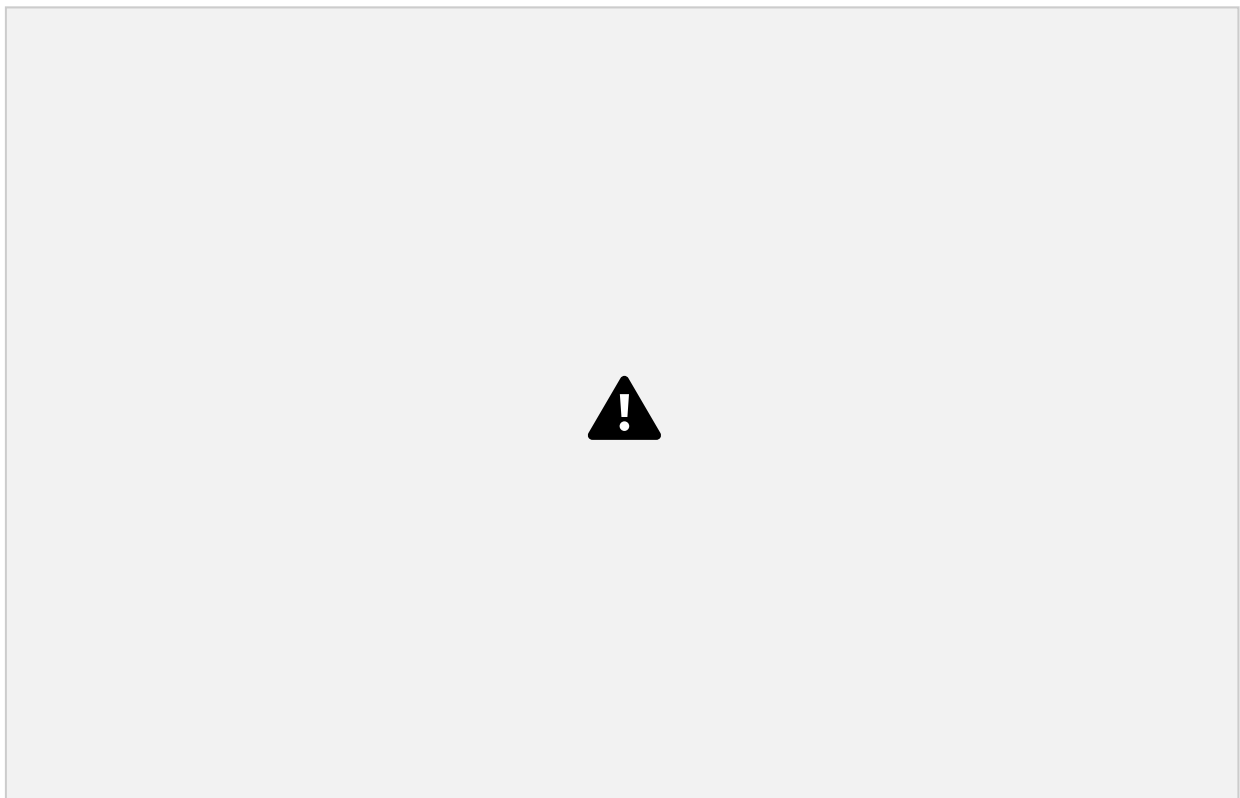


Now Click the "Accept the terms & download" Button





Once downloaded to your machine “**Run As an Administrator**” This Pop-up will appear before you begin the Wizard Check the **Agree terms and Install with Admin Privileges.**



Add a caption...

You will receive this pop-up as a Pre-check Click “**Close**” and begin the Wizard



Add a caption...

Checkmark this page to **connect your Active Directory Domain(s)** to prepare for Hybrid Setup. Click "**Next**"



Add a caption...

Now connect the wizard to your Microsoft Entra ID Cloud Instance with your Cloud Admin Account - Then Click “**Authenticate**” Using your 2-MFA then click “**Next**”



Add a caption...

Which gMSA do I choose? Custom or Standalone?



Add a caption...

In almost all modern Active Directory scenarios, a Group Managed Service Account (**gMSA**) is **better than a standalone** Managed Service Account (**sMSA**), primarily because it provides the same security benefits (automatic password management) but can be used across multiple servers.

sMSA (Standalone): Restricted to a single, specific computer account.

gMSA:(Custom): Designed for multiple servers, load balancers scenarios or server farms

For Our Needs sMSA (standalone) will do what we need for our Lab Environment.
Next We will grab the Connection Credentials from our on-premises Active Directory to provide the Tenants Last needed Information



Add a caption...

Lastly Confirm the Correct admin Credentials have been added to do the Agent provisioning on your machine.



Add a caption...

Now That Provisioning is done you can Press “Exit”



Add a caption...

Section 3: Performing a Microsoft Entra Cloud Sync

First within "M365 Entra Admin Center" under "Entra connect" Select “Cloud Sync Configuration” Once you've come to the Drop-Down Bar you will read - For A List of active Agents select “Click Here”



Add a caption...

You Will then Get a Tab on the Right side of your Browser to show your "Active" Configured Agent

Select your On-Premises Windows Active Directory Server



Add a caption...

In that the "**Sync Configurator**" has found your Windows Server **Agent**
Check mark the "**Enable Password hashing sync**" for Security and integrity for your
Users and Admins then **Click** the "**Create**" button to finish your Cloud sync setup.



Add a caption...

Now we can see our cloud sync from both On-premises to Entra ID - and Entra ID
to On-premises (AD) Status is **Healthy** and Ready to **Provision** its users



Add a caption...

You will go back to the “**Entra Connect**” Page “**Cloud Sync**”

A brand-new “**Cloud Sync Configuration Page**” will appear for your last few needed details you can select what "Scope" of Attributes and Object you want to Sync to your Entra Cloud First and some helpful **tutorial Videos** will be provided for your new Hybrid Setup before you Enable the Sync.

<https://aka.ms/VideoTutorial-CloudSyncOverview>

<https://aka.ms/VideoTutorial-CloudSyncFirstConfigSetup>

<https://aka.ms/VideoTutorial-CloudSyncAgentInstall>

<https://aka.ms/VideoTutorial-CloudSyncAccidentalDeletes>

<https://aka.ms/VideoTutorial-CloudSyncOnDemandProv>



Go to "**Review and Enable**" Click and Finish setup



Add a caption...

If you dont want to be bombarded with Notification emails set this up to be a " [no reply@contoso.com](mailto:reply@contoso.com)" If you need to monitor when the Entra Sync changes or has re connections.

nd

"Prevent accidental deletion do only a 2 confirmation can be made on the removal of a User Object.



Now how do we know if we are truly Hybrid-Joined?

We Run a Test within the Command Line to check the Hybrid Status of Our **Active Directory Domain Controller "DC01"**

-Open Command-Line as Administrator and Type the Following command
"dsregcmd /status"

AzureAdJoined = YES + DomainJoined = YES - "Microsoft-Entra Hybrid-Joined"



Add a caption...

Troubleshoot devices by using the dsregcmd command - Microsoft Entra ID

<https://learn.microsoft.com/en-us/entra/identity/devices/troubleshoot-device-dsregcmd>



Add a caption...

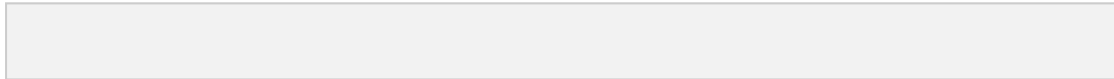


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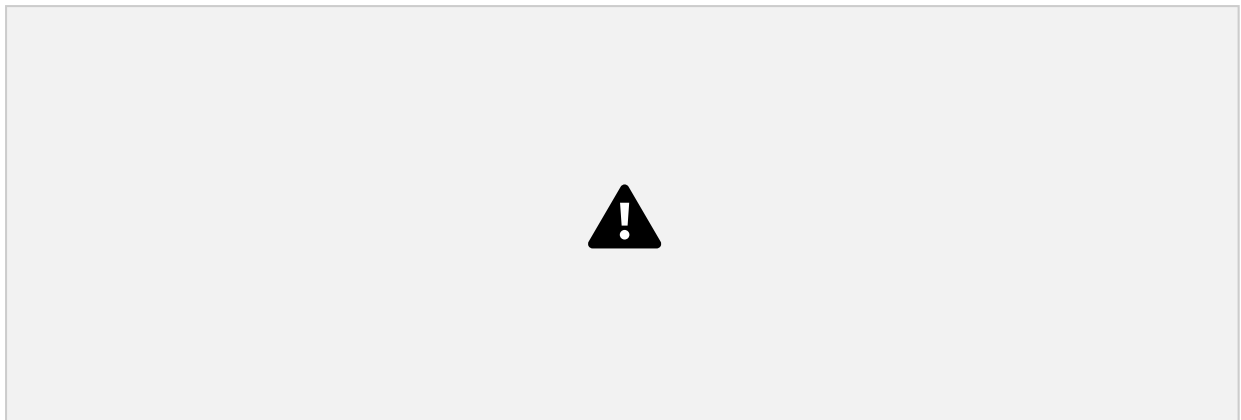
CONGRATS YOU ARE NOW HYBRID SYNCED!!!!

*For **Entra Connect Cloud Sync**, best practice for scoping is to use Organizational Unit (OU)-based filtering to restrict synchronization to only necessary user and group objects. This approach provides a lightweight footprint, allows for granular control over hybrid identities, and enables using multiple agents for high availability.

We Have a New User Named "**Jeremy**" In our "**ADUC**" Let perform a "**Provision Sync**" to Bring his User Object and Attributes to the Cloud In the "**Cloud Sync**" Page go to the Connection with the "**Sync Direction**" "**AD to Microsoft Entra ID**"

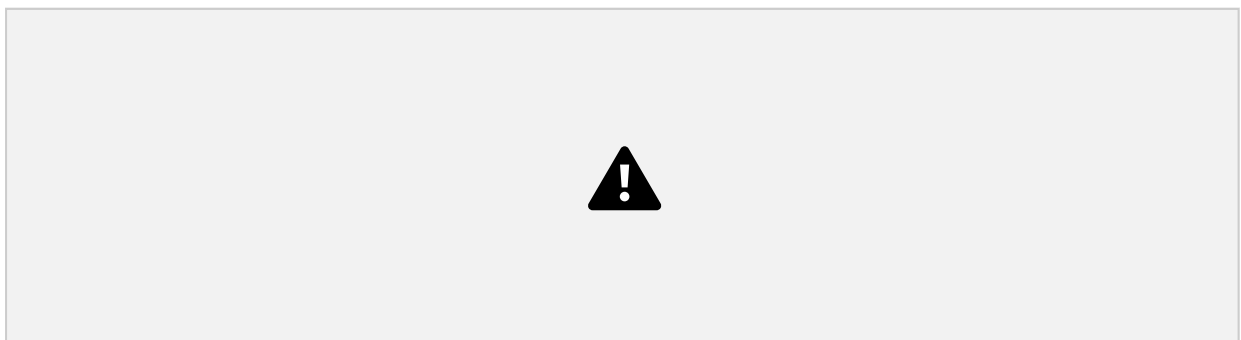


Add a caption...

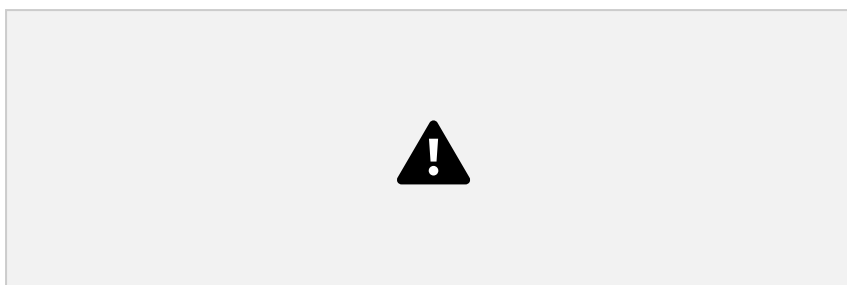


Add a caption...

Now Lets "Restart Provisioning" to sync the user



Add a caption...



Add a caption...Now we will find Jeremy within our "All Users" List within
Microsoft Entra Admin center - Users - All Users Scroll and Find his Name

CONGRATS YOU HAVE NOW HYBRID-SYNC YOUR USER FROM on-premises AD
to M365 Entra Users



Add a caption...



Add a caption...



Add a caption...

Step 4 Conclusion : Automatically register your Domain-Joined Computers

Have your newly Domain-joined computer be automatically added to Microsoft Entra ID Search in the taskbar "`gpedit.msc`" to get your "Group Policy Management Editor " on your AD server.



Enable the Group Policy **Register domain joined computers as devices** in **Computer**

Configuration > Policies > Administrative Templates > Windows Components >
Device Registration

add CNAME record for Validation "With your Domain "

[EnterpriseEnrollment.contoso.com](#) / [enterpriseenrollment-s.manage.microsoft.com](#)

[EnterpriseRegistration.contoso.com](#) / [EnterpriseRegistration.windows.net](#)



Verify CNAME

1. Sign in to the [Microsoft Intune admin center](#).
2. Go to **Devices**.

3. Expand **Device onboarding** and select **Enrollment**.
4. Select the **Windows** tab.
5. Under **Enrollment options**, select **CNAME Validation**.

For **Domain**, enter the company website, and then choose **Test**.



Additional Resources



<https://docs.azure.cn/en-us/entra/identity/hybrid/connect/how-to-connect-install-roadmap>

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment/deploy-hybrid>

<https://learn.microsoft.com/en-us/exchange/hybrid-deployment-prerequisites>

<https://learn.microsoft.com/en-us/microsoft-365/admin/setup/add-domain>

<https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/how-to-connect-install-prerequisites>